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Manuscript submitted by co-author Eko Arip Winanto on behalf of all authors.

To: The Editor-in-Chief
Journal of Information Security and Applications (Elsevier)

Re: Submission of an original research manuscript

Dear Editor,

We are pleased to submit our manuscript, **"QuantGhost: Deployment-Robust Weight Steganography and a Capacity–Robustness–Stealth Trilemma,"** for consideration as an original research article in the *Journal of Information Security and Applications* (JISA).

Pre-trained neural networks are increasingly shared through public model hubs, which turns the model file itself into a supply-chain attack surface. Prior weight-steganography attacks hide megabyte-scale payloads in float32 weights, but their survival under real deployment has never been tested. We show, on real models, that the routine transforms a production system applies—INT8 quantization, pruning, and fine-tuning—erase the mantissa-bit payloads of StegoNet- and EvilModel-style attacks entirely, and that a faithful spread-spectrum MaleficNet survives those transforms only at watermark-scale capacity. Our paper asks, and answers, whether a payload can survive the full deployment pipeline at high capacity while remaining stealthy and accuracy-preserving, and turns the answer into both a fundamental limit and a concrete defense.

The manuscript makes the following contributions:

- **A capacity–robustness–stealth trilemma.** For the function-preserving neuron-permutation group of convolutional and dense layers, we prove that no weight-embedding scheme can be simultaneously high-capacity, permutation-invariant, and stealthy-quantization-robust: any stealthy, quantization-robust, permutation-invariant channel is bounded by $\sum_\ell k_h^{(\ell)} k_w^{(\ell)}$ bits, a value our construction attains.
- **QuantGhost.** We present a quantization-aware, distribution-preserving embedding that occupies the high-capacity, stealthy, quantization-robust corner of the trilemma. It survives INT8, pruning, and fine-tuning at 3.2 MB on a real ResNet-50 for only 0.19 percentage points of accuracy, and survives byte-exactly through PyTorch's *production* INT8 kernel, not merely an analytical quantizer.
- **A measured MaleficNet baseline.** Rather than quote prior art from its reported design, we re-implement MaleficNet and run it on the same INT8/pruning/ fine-tuning pipeline. Its spread-spectrum payload does survive these transforms, but its recovery decays as $1/\sqrt{K}$ and reaches chance near one bit per weight, so it is transform-robust only at low capacity. This makes the trilemma concrete across two real constructions: only QuantGhost survives at high capacity.
- **An attack and its detector.** Against the released Model X-Ray steganalyzer, QuantGhost is statistically indistinguishable from benign weights on a diverse model population where the prior attack is detected. We propose a targeted sub-quantum residual detector that reliably flags the basic attack across a narrow and a diverse 3000-model zoo, while honestly showing that an adaptive attacker reaches chance on a realistic population, so

detection must be backed by a structural bound.

- **Limits of permutation defenses.** A faithful, full-model neuron permutation on a real ResNet-18, verified function-preserving to machine precision, removes the high-capacity payload; but a 196-bit permutation-invariant channel survives and bootstraps recovery of an out-of-band, AES-encrypted payload, so permutation-based sanitization is incomplete.
- **Generality and reproducibility.** We show the attack is architecture- and domain-agnostic on a real Vision Transformer and through a cross-dataset study on IoMT/IoT intrusion-detection models (CICIoMT, CICIoT), set out the requirements for a complete defense, and release the full code, data, and a one-to-one table/figure-to-script mapping as an open repository at github.com/robotumbel/quantghost.

We believe the work is a strong fit for JISA. The journal is a leading venue for information hiding, steganography and steganalysis, and machine-learning security; it has published closely related work, including the Model X-Ray steganalyzer that our attack evades and that we build our proposed detector against. Our manuscript contributes a proven capacity limit, a new high-capacity deployment-robust attack, a measured comparison with the strongest prior method, and a new detector, with direct relevance to the IoT/IoMT deployments JISA readers study.

This research is offensive in nature but carried out to strengthen defenses. All payloads in our experiments are encrypted random bytes or an AES key; no live malware is embedded, executed, or distributed. We will responsibly disclose our findings to the authors of the affected detector and defense (Model X-Ray and NeuPerm) prior to publication, and the paper proposes concrete mitigations.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submission and declare no competing interests.

Thank you for considering our work. We look forward to the reviewers' comments.

Sincerely,

Kurniabudi
Corresponding author, on behalf of all authors
(Eko Arip Winanto, Mohd Yazid Idris, Kurniabudi, and Febby Tri Ramadhanti)
Submitted via Editorial Manager by Eko Arip Winanto (winanto@graduate.utm.my).